COMPETENCY STANDARD

FOR

ADVANCED SEWING MACHINE OPERATION

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: https://sicip.gov.bd/

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (36 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-ASM-01-G	Apply Occupational Health and Safety (OHS) Practices at Workplace	1. Identify, control and report OHS hazards 2. Conduct work safely 3. Follow emergency response procedures 4. Maintain and improve health and safety in the workplace	18
SICIP-RMG-ASM -02-G	Work in a team environment	1. Identify team goals and work processes. 2. Communicate and co-operate with team members. 3. Work as a team member 4. Solve problems as a team member	18
		Total Hour	**36 Hrs.**

Sector Specific Competencies (45 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-ASM-01-S	Carry out measurements and calculations	1. Plan and prepare. 2. Obtain measurements. 3. Perform calculations.	09
SICIP-RMG-ASM-02-S	Interpret Sketch and Specifications in Manuals	1. Identify information from manual 2. Identify sketch and specifications	18
SICIP-RMG-ASM-03-S	Apply green practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	18
		Total Hour	**45 Hrs.**

Occupation Specific Competencies (279 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SEIP-RMG-ASM-01-O	Demonstrate Knowledge on Advanced Sewing Operation	1. Identify Advanced Sewing Machine and Process 2. Identify Parts and Functions of Advanced Sewing Machines 3. Identify Attachments of Advanced Sewing Machines	09
SEIP-RMG-ASM-02-O	Perform Advanced Single Needle Lock Stitch Machine Operation	1. Maintain OHS at Workplace 2. Prepare for Machine Operation 3. Operate Advanced Single Needle Lock Stitch Machine 4. Maintain Workplace Cleanliness and Restore Tools	81
SEIP-RMG-ASM-03-O	Perform Chain Stitch Machine Operation	1. Maintain OHS at Workplace 2. Prepare for Machine Operation 3. Operate Single Needle Chain Stich Machine 4. Operate Multi Needle Chain Stich Machine 5. Operate Advanced Overlock Machine 6. Operate Feed of the Arm Machine 7. Maintain Workplace Cleanliness and Restore Tools	90
SEIP-RMG-ASM-04-O	Perform Advanced Flat Lock Machine Operation	1. Maintain OHS at Workplace 2. Prepare for Machine Operation 3. Operate Advance Flat Lock Machine	18

Code	Unit of Competency	Elements of Competency	Duration (hours)
		4. Maintain Workplace Cleanliness and Restore Tools	
SEIP-RMG-ASM-05-O	Perform Auto Pocket Welting (APW) Machine Operation	1. Maintain OHS at Workplace 2. Prepare for Machine Operation 3. Operate Auto Pocket Welting (APW) Machine 4. Maintain Workplace Cleanliness and Restore Tools	27
SEIP-RMG-ASM-06-O	Perform Bar Tack Machine Operation	1. Maintain OHS at Workplace 2. Prepare for Machine Operation 3. Operate Bar Tack Machine 4. Maintain Workplace Cleanliness and Restore Tools	27
SEIP-RMG-ASM-07-O	Maintain Quality and Productivity Issues	1. Maintain Quality Procedures 2. Carry Out Basic Maintenance of Sewing Machine 3. Maintain Standard Productivity	27
		Total Hours	**279 Hrs.**

COMPETENCY STANDARD: ADVANCED SEWING MACHINE OPERATION

Generic Competencies:

Unit of Competency: **Apply Occupational Health and Safety (OHS) Practices at Workplace**	**Nominal Duration:** 18 Hrs**.**	**Unit Code:** SEIP-RMG-ASM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices at workplace. It specifically includes the tasks of identifying, controlling and report OSH hazards, conducting work safely, following emergency response procedures and maintaining and improving health and safety in the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify, control and report OHS hazards	1.1 Work area is routinely checked for Occupational Health and Safety (OHS) hazards prior to commencing and during work. 1.2 **Hazards** are identified and corrective action is taken within the level of responsibility. 1.3 OSH hazards and incidents are reported to appropriate personnel according to workplace procedures. 1.4 **Safety signs and symbols** are identified and followed. 1.5 **Preventive Controlmeasures** are identified in accordance with OSH work standards.
2. Conduct work safely	2.1 OHS practices are applied in the workplace. 2.2 **Personal Protective Equipment (PPE)** are used.
3. Follow emergency response procedures	3.1 Emergency situations are identified and reported according to workplace requirements. 3.2 Emergency procedures are followed as appropriate to the nature of the emergency and according to workplace procedures. 3.3 **Workplace procedures** for dealing with accidents, fires and emergencies are followed whenever necessary within scope of responsibilities.
4. Maintain and improve health and safety in the workplace	4.1 Risks are identified and appropriate control measures are implemented in the workplace. 4.2 Recommendations arising from risk assessments are implemented within level of responsibility.

	4.3 Safety records are maintained according to **<u>company policies</u>.**

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Hazards	1.1. OHS incidents include near misses, injuries, illnesses and property damage, noise, handling hazardous substances. 1.2. Biological hazards- bacteria, viruses, plants, parasites, mites, molds, fungi, insects 1.3. Chemical hazards – dusts, fibers, mists, fumes, smoke, gasses, vapors 1.4. Ergonomics Hazards 1.5. Physiological factors – monotony, personal relationship, work out cycle 1.6. Safety hazards (unsafe workplace condition) – confined space, excavations, falling objects, gas leaks, electrical, poor storage of materials and waste, spillage, waste and debris 1.7. Unsafe workers' act (Smoking in off-limited areas, Substance and alcohol abuse at work) 1.8. Working with and near moving equipment / load shifting equipment. 1.9. Broken or damaged equipment or materials.
2. Safety signs and symbols	2.1 Direction signs (exit, emergency exit, etc.) 2.2 First aid signs 2.3 Danger Tags 2.4 Hazard signs 2.5 Safety tags 2.6 Warning signs
3. Personal Protective Equipment (PPE)	3.1 Apron 3.2 Safety Helmet 3.3 Goggles 3.4 Ear muffs 3.5 Ear plugs 3.6 Gloves 3.7 Clothing 3.8 Safety Boots 3.9 Face Protection (Musk & Shield) 3.10 Scarf 3.11 Hair Net/cap/bonnet 3.12 Hard hat

4. Preventive Control Measures	4.1 Eliminate the hazard (i.e., get rid of the dangerous machine 4.2 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 4.3 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 4.4 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OSH-related topics, issue warning signage, rotation/shifting work schedule) 4.5 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 4.6 Use personal protective equipment 4.7 Safety, Health and Work Environment Evaluation 4.8 Periodic and/or special medical examinations of workers
5. Workplace Procedures	5.1 Fire fighting 5.2 Medical and first aid 5.3 Evacuation 5.4 Material Safety Data Sheets (MSDSs) and manufacturers' advice
6. Company policies	6.1 Job related Standard Operating Procedures (SOPs). 6.2 Occupational Health and Safety (OSH) specific procedures.

Curricular Content Guide

1. Underpinning Knowledge	1.1. OSH Workplace Policies and Procedures. 1.2. Work safety procedures. 1.3. Fire and emergency procedures. 1.4. Types of Hazards (Biological, Chemical and Physical) and their effects. 1.5. PPE types and uses. 1.6. Personal hygiene practices. 1.7. OSH awareness. 1.8. Steps of hazard identification. 1.9. Principles of hazards control. 1.10. Employer's role. 1.11. Supervisor's responsibilities.
2. Underpinning Skills	2.1. Identifying OHS policies and procedures. 2.2. Following personal work safety practices. 2.3. Reporting hazards and risks. 2.4. Responding to emergency procedures.

	2.5. Maintaining physical well-being in the workplace. 2.6. Identify tools and equipment related to OHS. 2.7. Improving OHS performance.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Manuals 4.4 Workplace documents, signs and symbols 4.5 Relevant specifications or work instructions. 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 used Personal Protective Equipment (PPE). 1.2 identified hazards. 1.3 took corrective action of different hazards. 1.4 took corrective action for emergency procedure. 1.5 reported emergency situation to the Supervisor / Manager. 1.6 satisfied requirements mentioned in the performance criteria and range of variables.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:** 18 Hrs**.**	**Unit Code:** SEIP-RMG-ASM-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work in a team environment. It specifically includes the tasks of Identifying team goals and processes, communicating and cooperate with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes.	1.1. Team goals and collaborative decision-making processes are identified. 1.2. Role and common goals of the team are defined from available **sources of information.** 1.3. Team structure, responsibilities and reporting relations are identified from team discussions and other external sources.
2. Communicate and co-operate with team members.	2.1 Communication and negotiation skills are applied and maintained in all relevant situations. 2.2 Constructive contributions are made to **workplace discussions** on such issues as production, quality and safety. 2.3 Goals/ objectives and action plans undertaken in the workplace are communicated promptly. 2.4 Information regarding problems and issues are organized coherently to ensure clear and effective communication. 2.5 Dialogue is initiated with appropriate personnel. 2.6 Communication problems and issues are raised 2.7 Barriers to communication are identified and resolved
3. Work as a team member	3.1 Effective forms of communication are used to interact with **team members** in discussing team activities and objectives. 3.2 Mutual respect, empathy and active collaboration are demonstrated 3.3 Communication channels are followed as per **workplace context.**
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified

	4.2 Problems are investigated and analyzed 4.3 Potential solutions of problem are identified 4.4 Recommendations about possible solutions are developed, documented, ranked and presented to team members for decision.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Sources of information	1.1. Organizational structures 1.2. Operations Manuals 1.3. Job description 1.4. Standard operating procedures
2. Workplace discussions	2.1 Coordination meetings 2.2 Toolbox discussion 2.3 Peer-to-peer discussion
3. Team members	3.1 Coach / members 3.2 Supervisor / manager 3.3 Peers / colleagues 3.4 Other members /Employee representative of the organization.
4. Workplace context	4.1 National Laws and Statutes 4.2 Standard Operating Procedures 4.3 Workplace Rules and Regulations

Curricular Content Guide

1. Underpinning Knowledge	1.1. Sources of information define 1.2. Team structure, role, and responsibility. 1.3. Individual member's roles and responsibilities. 1.4. Effective verbal communication methods 1.5. Communication flow and reporting structures. 1.6. Interpersonal communication skills. 1.7. Organization requirements for written and electronic communication methods 1.8. Communication problems and issues 1.9. Barriers in communication 1.10. Team planning. 1.11. Team meeting procedures. 1.12. Workplace etiquette 1.13. Industry maintenance, service and helpdesk practices, processes and procedures 1.14. Industry standard diagnostic tools 1.15. Malfunctions and resolutions

2. Underpinning Skills	2.1 Organizing sources of information 2.2 Identifying the role and responsibility of the team. 2.3 Identifying roles and responsibilities of individual members. 2.4 Identifying effective verbal communication methods 2.5 Identifying communication flow and reporting structure. 2.6 Identifying interpersonal communication skills 2.7 Complying with organization requirements for the use of written and electronic communication methods 2.8 Negotiation and communication skills 2.9 Participating in team discussion. 2.10 Working as a team member. 2.11 Participating in a variety of workplace discussions 2.12 Effective clarifying and probing skills 2.13 Identifying issues 2.14 Identifying current industry standard diagnostic tools 2.15 Describing common malfunctions and resolutions. 2.16 Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.3 communicated and cooperated with team members 1.4 practice problem solving within the team

2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Sector-Specific Competencies

Unit of Competency: **CARRY OUT MEASUREMENTS AND CALCULATIONS**	**Nominal Duration:** 09 hrs.	**Unit Code:** SICIP-RMG-ASM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for carry out measurements and calculations. It specifically includes the task of planning and preparing, obtaining measurements and performing calculations.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan and prepare	1.1. Work instructions are confirmed and applied to the job in hand. 1.2. Materials to be measured are identified as per job specification. 1.3. Appropriate **measuring device** is identified and selected based on materials to be measured. 1.4. Specifications are obtained and verified from relevant **documents.**
2. Obtain measurements	2.1 Method of obtaining measurement is selected and applied. 2.2 **Measurements** are obtained using appropriate device in accordance with job requirement. 2.3 Measurements, including area, volume, tolerance and clearance limits, are confirmed and applied
3. Perform calculations	3.1 **Calculations**, using basic operations, for determining material requirement are taken. 3.2 Appropriate **formulas** for calculating quantities are selected. 3.3 Quantities are estimated from the calculation taken. 3.4 Material quantities are calculated, confirmed and recorded within tolerances.

Range of Variables

Variable	Range May include but not limited to:
1. Measuring device	1.1 Measuring tape 1.2 Steel rule 1.3 Calculator 1.4 Sets square
2. Documents	2.1. Technical manuals 2.2. Specifications 2.3. Sketches 2.4. Drawings 2.5. Charts 2.6. Photographs
3. Measurements	3.1. Length 3.2. Width 3.3. Weight 3.4. Tolerance
4. Calculations	4.1 Addition 4.2 Subtraction 4.3 Multiplication 4.4 Division 4.5 Area 4.6 Volume 4.7 Circumference 4.8 Cubic meter 4.9 Volumetric weight
5. Formulas	5.1 Fractions 5.2 Percentages 5.3 Mixed numbers 5.4 Conversions 5.5 Scales

Curricular Content Guide

1. Underpinning Knowledge	1.1 Measuring devices 1.2 Measurements 1.3 Simple calculation techniques 1.4 Appropriate formulas for calculating quantities 1.5 Estimating quantities
2. Underpinning Skills	2.1 Identifying appropriate measuring devices 2.2 Carrying out measurements 2.3 Performing calculations using appropriate formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate measuring devices 1.2 carried out measurements for appropriate device 1.3 identified and selected correct mathematical formula 1.4 performed calculations using appropriate formulas
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET SKETCH AND SPECIFICATIONS IN MANUALS**	**Nominal Duration:** 18 Hrs**.**	**Unit Code:** SEIP-RMG-ASM-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret sketch and specifications in manuals. It specifically includes the tasks of identifying information from manual and identifying sketch and specifications.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify information from manual	1.1. Appropriate **manuals** are collected as per sample. 1.2. Importance of manuals is recognized. 1.3. Required information are collected from manuals.
2. Identify sketch and specifications	2.1 Relevant **sketch** and **specifications** are identified. 2.2 Key **terms and abbreviations** are identified. 2.3 **Signs and symbols** are identified. 2.4 Schedules, dimensions, drawings and specifications are interpreted.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Manuals	1.1 Buyer's specification manual 1.2 Compliance manual 1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Signs and symbols, instruction manuals
2. Sketch	2.1 Technical sketch 2.2 Measurement sketch
3. Specifications	3.1 Product specifications 3.2 Performance specifications 3.3 Method specifications
4. Terms and abbreviations	4.1 Refers to all terms and abbreviations associated with the RMG sector
5. Signs and symbols	5.1 Include all signs and symbols associated with the RMG sector

Curricular Content Guide

1. Underpinning Knowledge	1.1 Themes on various types of RMG manuals 1.2 Units of measurement 1.3 Units of conversion 1.4 Rules of sketch, drawings and specifications
2. Underpinning Skills	2.1 Recognising importance of manual 2.2 Selecting appropriate manuals as per sample 2.3 Collecting information from manual as per sample 2.4 Interpreting schedules, dimensions, drawings and specifications
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected information from manual as per sample 1.2 identified sketches and specifications as per sample
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN RMG SECTOR**	**Nominal Duration:** 18 hrs.	**Unit Code:** SICIP-RMG-ASM-03-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **Principles of green practices in RMG (Ready-Made Garment)** manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 **Sources of environmental impacts** during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 **Ways of mitigating environmental impacts** in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 **Recyclable and non-recyclable items** in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 **Different types of waste in RMG manufacturing** are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range May include but not limited to:
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 **Recyclable Items in RMG Manufacturing** 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers.

	4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 **Non-Recyclable Items in RMG Manufacturing** 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in RMG manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in RMG manufacturing 2.6 Disposing hazardous waste in RMG production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-Specific Competencies (279 Hrs.)

Unit of Competency: DEMONSTRATE KNOWLEDGE ON ADVANCED SEWING OPERATION	Nominal Duration: 09 Hrs.	Unit Code: SEIP-RMG-ASM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to demonstrate knowledge on advanced sewing operation. It specifically includes the tasks of identifying advanced sewing machine and process, identifying parts and functions of advanced sewing machines and identifying attachments of advanced sewing machines.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify advanced sewing machine and process	1.1 **Advanced sewing machines** are identified as per job requirement. 1.2 **Single needle lock stitch machine operation** is identified as per job requirement 1.3 **Double needle lock stitch machine operation** is identified as per job requirement 1.4 **Overlock machine operation** is identified as per job requirement 1.5 **Feed of the arm machine operation** is identified as per job requirement 1.6 **Multi needle chain stitch machine operation** is identified as per job requirement. 1.7 **Flat lock machine operation** is identified as per job requirement
2. Identify parts and functions of advanced sewing machines	2.1 **Single needle lock stitch machine parts** are identified. 2.2 **Overlock machine parts** are identified. 2.3 **Feed of the arm machine parts** are identified. 2.4 **Multi needle chain stitch machine parts** are identified. 2.5 **Flat lock machine parts** are identified. 2.6 Functions of advance sewing machine parts are identified
3. Identify attachments of advanced sewing machines	3.1 **Attachments of advanced sewing machines** are identified. 3.2 Functions of attachments of advanced sewing machines are described.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Advanced sewing machines	1.1 Single needle lock stitch machine (UBT) 1.2 Double needle lock stitch machine 1.3 Overlock machine 1.4 Feed of the arm 1.5 Multi needle chain stitch machine 1.6 Flat lock machine 1.7 Auto Pocket Welt (APW) 1.8 Auto Pocket Setter (APS) 1.9 Bar tack machine 1.10 Zigzag Machine 1.11 Blind Stitch Machine 1.12 Pattern Sewer Machine 1.13 Flat Seamer Machine
2. Single needle lock stitch machine operation	2.1 Collar band joint 2.2 Pocket attach/topstitch 2.3 Placket attaches 2.4 Box making 2.5 Label attachment 2.6 Bottom hem 2.7 Waist band mouth close 2.8 Back neck piping top stitch 2.9 Welt pocket making 2.10 Cuff join/top stitch 2.11 Epilate make 2.12 Sleeve Placket join (shirt) 2.13 Side vent join 2.14 Armhole topstitches 2.15 Zipper join
3. Double needle lock stitch machine operation	3.1 Zipper join 3.2 Front pocket opening topstitches 3.3 Button placket makes 3.4 Flap make
4. Overlock machine operation	4.1 Neck join 4.2 Sleeve join 4.3 Side seam 4.4 Back yoke/rise join 4.5 Waist Belt join 4.6 Cuff join 4.7 Collar join
5. Feed of the arm machine operation	5.1 Back rise/yoke join 5.2 Side/in seam join/top stitch

	5.3 Shoulder to shoulder top stitch
6. Multi needle chain stitch machine operation	6.1 Waist band join 6.2 Make box placket (shirt) 6.3 Waist band top stitch 6.4 Smocking stitching
7. Flat lock machine operation	7.1 Round sleeve hem 7.2 Bottom hem 7.3 Top stitches 7.4 Facing join
8. Single needle lock stitch machine parts	8.1 Feed dog 8.2 Presser foot 8.3 Needle plate 8.4 Bobbin winder 8.5 Needle guard 8.6 Slide plate 8.7 Thread take-up lever 8.8 Presser foot lifter 8.9 Tension spring 8.10 Tension disc 8.11 Stitch (SPI) regulator 8.12 Back stitch lever 8.13 Hand wheel 8.14 Bobbin 8.15 Bobbin case 8.16 Thread spool tray 8.17 Electronic display board
9. Overlock machine parts	9.1 Presser foot 9.2 Needle plate 9.3 Side cover 9.4 Looper cover 9.5 Balance wheel 9.6 Power cord attachment 9.7 Stitch length dial 9.8 Telescopic thread guide 9.9 Lower looper thread tension dial 9.10 Right needle thread tension dial 9.11 Left needle thread tension dial
10. Feed of the arm machine parts	10.1 Thread guide 10.2 Thread tensioner 10.3 Thread take-up lever 10.4 Needle 10.5 Folder (lapped) 10.6 Feed dog 10.7 Throat plate

	10.8 Looper 10.9 Presser foot bar 10.10 Presser foot 10.11 Needle plate throat plate 10.12 Presser foot lever 10.13 Machine pulley 10.14 Front cover 10.15 Needle bar 10.16 Upper thread guide
11. Multi needle chain stitch machine parts	11.1 Thread stands 11.2 Thread guide 11.3 Hand wheel 11.4 Thread tensioner 11.5 Needle clamp 11.6 Tension plate 11.7 Looper thread tension guide
12. Flat lock machine parts	12.1 Thread stands 12.2 Thread guides 12.3 Disc type tensioner 12.4 Pressure feed lever 12.5 Thread take-up lever 12.6 Needle 12.7 Looper
13. Attachments of advanced sewing machines	13.1 LED/Laser Light 13.2 Welted folder 13.3 Double folder 13.4 CL and CR guide 13.5 Plain feed 13.6 Shearing guide 13.7 D-set top stitch feed 13.8 Magnet guide 13.9 T-guide 13.10 Spring guide 13.11 Zig and template

Curricular Content Guide

1. Underpinning Knowledge	1.1 Sewing machines operation procedure 1.2 Functions of the sewing machines 1.3 Different parts of sewing machines 1.4 Functions of the different parts of sewing machines 1.5 Describe sewing attachments 1.6 Functions of attachments
2. Underpinning Skills	2.1 Following OHS 2.2 Listing advanced sewing machine

	2.3 Listing advanced sewing machine process 2.4 Listing advanced sewing machine parts 2.5 Explaining different types sewing machine functions 2.6 Using sewing attachments
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified advanced sewing machine and process 1.2 identified parts and function of advanced sewing machine 1.3 identified attachments of advanced sewing machines
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM ADVANCED SINGLE NEEDLE LOCK STITCH MACHINE OPERATION**	**Nominal Duration:** 81 Hrs.	**Unit Code:** SEIP-RMG-ASM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform advanced single needle lock stitch machine operation. It specifically includes the tasks of maintaining OHS at workplace, preparing for machine operation, operating advanced single needle lock stitch machine and maintaining workplace cleanliness and restore tools.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain OHS at workplace	1.1 **Personal Protective Equipment (PPE)** is collected and used as per job requirement. 1.2 Occupational Health and Safety (OHS) are followed as per job requirement.
2. Prepare for machine operation	2.1 **Tools** are identified and checked for proper operating condition 2.2 **Materials** are collected and checked in accordance with design/style requirement 2.3 Required size and **type of needle** is identified in accordance with fabric and job requirements.
3. Operate advanced single needle lock stitch machine	3.1 Machine settings and tensions are adjusted as instructed in the operation manual. 3.2 Machine threading is completed as per standard. 3.3 Fabric is aligned and positioned correctly under the presser foot. 3.4 **Sewing operations** are carried out smoothly while ensuring consistent stitching quality. 3.5 **Common operational issues** are recognized and addressed using standard troubleshooting steps.
4. Maintain workplace cleanliness and restore tools	4.1 Workplace area is inspected and cleaned following established safety and hygiene procedures. 4.2 Waste materials are collected, segregated, and disposed of according to workplace guidelines. 4.3 Tools and equipment used are cleaned, checked for damage, and restored to their designated storage locations.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1 Face mask 1.2 Apron 1.3 Scarf/cap 1.4 Needle guard
2. Tools	2.1 Measurement tape 2.2 Scissors 2.3 Approved finish pattern 2.4 Marking pen/chalk 2.5 Brush 2.6 Bin
3. Materials	3.1 Fabric 3.2 Thread 3.3 Trims and Accessories
4. Type of needle	4.1 Ball point 4.2 Sharp point 4.3 Universal
5. Sewing operations	5.1 Sew straight seams 5.2 Sew corner seams (turning technique) 5.3 Sew curves (inward/outward) 5.4 Backstitching to secure seam ends 5.5 Edge stitching (close to edge) 5.6 Topstitching (decorative/functional stitch) 5.7 Seam joining of multiple layers
6. Common operational issue	6.1 Unusual machine noise 6.2 Excess oil/lubricant 6.3 Unusual needle breakage 6.4 Irregular stitch tension

Curricular Content Guide

1. Underpinning Knowledge	1.1 Required tools and materials 1.2 Types of needles 1.3 Sewing operation procedures 1.4 Machine threading procedures 1.5 Machine adjustments 1.6 Characteristics of fabrics and fabric alignment procedures 1.7 Unusual operating condition 1.8 Waste materials disposing procedure
2. Underpinning Skills	2.1 Following OSH 2.2 Checking and adjusting sewing machine

	2.3 Applying techniques of threading 2.4 Operating single needle lock stitch machine 2.5 Carrying out sewing operation 2.6 Checking and reporting unusual operating condition 2.7 Cleaning workplace and disposing waste materials
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 maintained OSH at workplace 1.2 collected tools and materials for job requirements 1.3 operated advanced single needle lock stitch machine 1.4 maintained workplace cleanliness and restored tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM CHAIN STITCH MACHINE OPERATION**	**Nominal Duration:** 90 Hrs.	**Unit Code:** SEIP-RMG-ASM-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform chain stitch machine operation. It specifically includes the tasks of maintaining OHS at workplace, preparing for machine operation, operating single needle chain stich machine, operating multi needle chain stich machine, operating advanced overlock machine, operating feed of the arm machine and maintaining workplace cleanliness and restore tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain OHS at workplace	1.1 **Personal protective equipment (PPE)** is collected and used as per job requirement. 1.2 Occupational health and safety and (OHS) are followed as per job requirement.
2. Prepare for machine operation	2.1. **Tools** are identified and checked for proper operating condition 2.2. **Materials** are collected and checked in accordance with design/style requirement 2.3. Required size and **type of needle** is identified in accordance with fabric and job requirements.
3. Operate single needle chain stich machine	3.1 Machine is threaded correctly as per manufacturer's instruction. 3.2 Machine speed and tension are adjusted to ensure smooth operation. 3.3 Fabric is positioned and guided under the needle according to stitching requirements. 3.4 **Chain stitch operations** are carried out maintaining uniform stitch formation and quality. 3.5 **Common operational issues** are recognized and addressed using standard troubleshooting steps.
4. Operate multi needle chain stich machine	4.1 Threads are inserted and adjusted through multiple needles and loopers as per threading diagram. 4.2 Required attachments are adjusted based on job requirements. 4.3 Fabric is positioned and aligned under the needle. 4.4 **Multi-needle chain stitching operation** is performed ensuring even stitch formation and fabric feeding.

	4.5 Common operational issues are recognized and addressed using standard troubleshooting steps.
5. Operate advanced overlock machine	5.1 Machine is threaded correctly following standard threading diagrams for upper and lower loopers and needles. 5.2 Required attachments are adjusted based on job requirements. 5.3 Fabric edges are aligned and positioned under the presser foot as per stitching specifications. 5.4 **Overlock stitching** is carried out maintaining even seam formation and consistent stitch quality. 5.5 Common operational issues are recognized and addressed using standard troubleshooting steps.
6. Operate feed of the arm machine	6.1 Machine is threaded and adjusted as per the manufacturer's guidelines. 6.2 Machine settings tension, and stitch length are adjusted to ensure smooth operation. 6.3 Two parts of fabric edge are positioned in folder as per requirement. 6.4 **Stitching operations** are performed maintaining consistent seam quality and proper fabric feeding. 6.5 Common operational issues are recognized and addressed using standard troubleshooting steps.
7. Maintain workplace cleanliness and restore tools	7.1 Workplace area is inspected and cleaned following established safety and hygiene procedures. 7.2 Waste materials are collected, segregated, and disposed of according to workplace guidelines. 7.3 Tools and equipment used are cleaned, checked for damage, and restored to their designated storage locations.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1 Face mask 1.2 Apron 1.3 Scarf/cap 1.4 Needle guard
2. Tools	2.1 Measurement tape 2.2 Scissors 2.3 Approved finish pattern 2.4 Folder or attachment 2.5 Marking pen/chalk 2.6 Brush

	2.7 Bin
3. Materials	3.1 Fabric 3.2 Thread 3.3 Trims and Accessories
4. Type of needle	4.1 Ball point 4.2 Sharp point 4.3 Universal
5. Chain stitch operation	5.1 Sew straight seams 5.2 Sew corner seams (turning technique) 5.3 Sew curves (inward/outward) 5.4 Topstitching (decorative/functional stitch) 5.5 Seam joining of multiple layers
6. Common operational issue	6.1 Unusual machine noise 6.2 Excess oil/lubricant 6.3 Unusual needle breakage 6.4 Irregular (skipped) stitch
7. Multi-needle chain stitching operation	7.1 Sew straight seams 7.2 Topstitching (multiple layers) 7.3 Waist band or box placket join
8. Stitching operations	8.1 Back yoke join 8.2 Back rise joins 8.3 In seam join 8.4 Side seam top stitch

Curricular Content Guide

1. Underpinning Knowledge	1.1 OHS procedure 1.2 Required tools and materials 1.3 Size and type of needle 1.4 Sewing operation procedures 1.5 Machines threading procedures 1.6 Threading procedure 1.7 Unusual operating condition of machines 1.8 Machines adjustment 1.9 Quality procedures
2. Underpinning Skills	2.1 Following OSH 2.2 Checking proper operating condition of machines 2.3 Identifying and collecting tools and materials 2.4 Completing machine threading 2.5 Operating single needle chain stitch machine 2.6 Operating multi needle chain stitch machine 2.7 Operating advanced overlock machine 2.8 Operating feed of the arm machine 2.9 Carrying out sewing operation

	2.10 Checking unusual operating condition 2.11 Maintaining stitch quality 2.12 Maintaining workplace cleanliness
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 maintained OHS at workplace 1.2 prepared for machine operation 1.3 operated single needle chain stitch machine 1.4 operated multi needle chain stitch machine 1.5 operated advanced overlock machine 1.6 operated feed of the arm machine 1.7 performed stitch operations 1.8 maintained workplace cleanliness and restored tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM ADVANCED FLAT LOCK MACHINE OPERATION**	**Nominal Duration:** 18 Hrs**.**	**Unit Code:** SEIP-RMG-ASM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform advanced flat lock machine operation. It specifically includes the tasks of maintaining OHS at workplace, preparing for machine operation, operating advance flat lock machine and maintaining workplace cleanliness and restore tools.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Maintain OHS at workplace	1.1 **Personal protective equipment (PPE)** is collected and used as per job requirement. 1.2 Occupational health safety (OHS) is followed as per job requirement.
2. Prepare for machine operation	2.1. **Tools** are identified and checked for proper operating condition 2.2. **Materials** are collected and checked in accordance with design/style requirement 2.3. Required size and **type of needle** is identified in accordance with fabric and job requirements
3. Operate advance flat lock machine	3.1 Materials are organized and set as per job specifications. 3.2 Machine is threaded including upper and lower loopers and needles, according to threading procedure. 3.3 Fabric edges are positioned and aligned properly on the flatbed for stitching. 3.4 **Flat lock stitching** is carried out ensuring uniform seam formation and stitch tension. 3.5 Unusual condition of machine is checked and reported to authority
4. Maintain workplace cleanliness and restore tools	4.1 Workplace area is inspected and cleaned following established safety and hygiene procedures. 4.2 Waste materials are collected, segregated, and disposed of according to workplace guidelines. 4.3 Tools and equipment used are cleaned, checked for damage, and restored to their designated storage locations.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1 Face mask 1.2 Apron 1.3 Scarf/cap 1.4 Needle guard
2. Tools	2.1 Measurement tape 2.2 Scissors 2.3 Approved finish pattern 2.4 Folder or attachment 2.5 Marking pen/chalk 2.6 Brush 2.7 Bin
3. Materials	3.1 Fabric 3.2 Thread 3.3 Trims and Accessories
4. Type of needle	4.1 Ball point 4.2 Sharp point 4.3 Universal
5. Flat lock stitching	5.1 Facing join 5.2 Loop makes

Curricular Content Guide

1. Underpinning Knowledge	1.1 OHS procedures 1.2 Machine parts and needle 1.3 Advance flat look machine operation procedure 1.4 Threading procedures 1.5 Garments faults and rectifying procedures 1.6 Flat lock stitching procedures 1.7 Unusual operating condition of machines 1.8 Workplace cleaning procedures
2. Underpinning Skills	2.1 Checking and cleaning machine 2.2 Checking tools and materials 2.3 Checking machine parts 2.4 Checking tension of the stitches 2.5 Carrying out flat lock stitching 2.6 Checking and reporting unusual condition 2.7 Maintaining workplace cleanliness
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn

	3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Maintained OHS at workplace 1.2 Prepared for machine operation 1.3 Operated advanced flat lock machine 1.4 Carried out flat lock stitching ensuring uniform seam formation and stitch tension. 1.5 Maintained workplace cleanliness and restored tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM AUTO POCKET WELTING (APW) MACHINE OPERATION**	**Nominal Duration:** 27 Hrs.	**Unit Code:** SEIP-RMG-ASM-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform auto pocket welting (APW) machine operation. It specifically includes the tasks of maintaining OHS at workplace, preparing for machine operation, operating auto pocket welting (APW) machine and maintaining workplace cleanliness and restore tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain OHS at workplace	1.1 **Personal protective equipment (PPE)** is collected and used as per job requirement. 1.2 Occupational health and safety and (OHS) are followed as per job requirement.
2. Prepare for machine operation	2.1 **Tools** are identified and checked for proper operating condition 2.2 Materials are collected and checked in accordance with design/style requirement 2.3 Required size and **type of needle** is identified in accordance with fabric and job requirements
3. Operate auto pocket welting (APW) machine	3.1 Back panel and pocketing are collected according to job specifications. 3.2 The machine is threaded and set up following the standard operating instructions. 3.3 Back panel and pocketing are aligned and positioned under the template or clamp. 3.4 Welt pocket is carried out according to the design. 3.5 Welt pocket is checked to ensure pocket mouth size. 3.6 Unusual condition of machine is checked and reported to authority.
4. Maintain workplace cleanliness and restore tools	4.1 Workplace area is inspected and cleaned following established safety and hygiene procedures. 4.2 Waste materials are collected, segregated, and disposed of according to workplace guidelines. 4.3 Tools and equipment used are cleaned, checked for damage, and restored to their designated storage locations.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1 Face mask 1.2 Apron 1.3 Scarf/cap 1.4 Needle guard
2. Tools	2.1 Measurement tape 2.2 Scissors 2.3 Approved finish pattern 2.4 Folder or attachment 2.5 Marking pen/chalk 2.6 Brush 2.7 Bin
3. Type of needle	3.1 Ball point 3.2 Sharp point 3.3 Universal

Curricular Content Guide

1. Underpinning Knowledge	1.1 OHS procedures 1.2 Required tools and materials 1.3 Machine parts and needle 1.4 Threading procedures 1.5 Auto pocket welting (APW) machine operation procedures 1.6 Unusual operating condition of machines 1.7 Cleaning procedures of tools and equipment 1.8 Waste materials disposal procedure
2. Underpinning Skills	2.1 Checking and cleaning machine 2.2 Checking tools and materials 2.3 Checking machine parts 2.4 Checking tension of the stitches 2.5 Carried out welt pocket. 2.6 Maintaining quality 2.7 Checking and reporting unusual condition 2.8 Maintaining workplace cleanliness
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the

	workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 maintained OSH at workplace 1.2 prepared for machine operation 1.3 identified rrequired size and type of needle 1.4 operated auto pocket welting (APW) machine 1.5 maintained workplace cleanliness and restored tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM BAR TACK MACHINE OPERATION**	**Nominal Duration:** 27 Hrs**.**	**Unit Code:** SEIP-RMG-ASM-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform bar tack machine operation. It specifically includes the tasks of maintaining OHS at workplace, preparing for machine operation, operating bar tack machine and maintaining workplace cleanliness and restore tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain OHS at workplace	1.1 **Personal protective equipment (PPE)** is collected and used as per job requirement. 1.2 Occupational health and safety (OHS) are followed as per job requirement.
2. Prepare for machine operation	2.1 **Tools** are identified and checked for proper operating condition 2.2 **Materials** are collected and checked in accordance with design/style requirement 2.3 Required size and type of needle is identified in accordance with fabric and job requirements
3. Operate bar tack machine	3.1 Machine is threaded following standard threading procedures. 3.2 Machine settings are adjusted to meet production specifications. 3.3 Garments are positioned under the machine's presser foot for bar tack stitching. 3.4 **Bar tack operations** are carried out ensuring uniform stitch density and reinforcement strength. 3.5 Stitch quality is checked as per quality standards. 3.6 Unusual condition of machine is checked and reported to authority.
4. Maintain workplace cleanliness and restore tools	4.1 Workplace area is inspected and cleaned following established safety and hygiene procedures. 4.2 Waste materials are collected, segregated, and disposed of according to workplace guidelines. 4.3 Tools and equipment used are cleaned, checked for damage, and restored to their designated storage locations.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1 Face mask 1.2 Apron 1.3 Scarf/cap 1.4 Needle guard
2. Tools	2.1 Measurement tape 2.2 Scissors 2.3 Approved finish pattern 2.4 Marking pen/chalk 2.5 Brush 2.6 Bin
3. Type of needle	3.1 Ball point 3.2 Sharp point 3.3 Universal
4. Bar tack operations	4.1 Belt loop 4.2 Pocket corners 4.3 Fly end 4.4 Collar points 4.5 Placket ends

Curricular Content Guide

1. Underpinning Knowledge	1.1 OHS procedures 1.2 Required tools and materials 1.3 Machine parts and needle 1.4 Threading procedures 1.5 Bar tack machine operation procedures 1.6 Stitch quality 1.7 Unusual operating condition of machines 1.8 Cleaning procedures of tools and equipment 1.9 Waste materials disposal procedure
2. Underpinning Skills	2.1 Checking and cleaning machine 2.2 Checking tools and materials 2.3 Checking machine parts 2.4 Checking tension of the stitches 2.5 Operating bar tack machine 2.6 Maintaining quality 2.7 Checking and reporting unusual condition 2.8 Cleaning workplace and tools and equipment
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 maintained OHS at workplace 1.2 prepared for machine operation 1.3 adjusted machine settings to meet production specifications. 1.4 operated Bar tack machine 1.5 maintained workplace cleanliness and restored tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MAINTAIN QUALITY AND PRODUCTIVITY ISSUES**	**Nominal Duration:** 27 Hrs**.**	**Unit Code:** SEIP-RMG-ASM-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to maintain quality and productivity issues. It specifically includes the tasks of maintaining quality procedures, carrying out basic maintenance of sewing machine and maintaining standard productivity.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain quality procedures	1.1 Quality specifications are interpreted as per job requirement. 1.2 Quality standard is followed as per mock-up. 1.3 Produced garments are inspected by self. 1.4 **Sewing defects** are identified. 1.5 Faults are rectified as per standard procedure.
2. Carry out basic maintenance of sewing machine	2.1 **Basic maintenance of sewing machine** requirements is identified and interpreted 2.2 Basic maintenance is performed as per job requirement 2.3 Machine is cleaned and lubricated as per machine manual.
3. Maintain standard productivity	3.1 Productivity targets and standards are understood. 3.2 Work schedules and plans are understood as per production requirements and timelines. 3.3 **Productivity improvement techniques** are identified to optimize work processes and productivity.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Sewing defects	1.1 Seam puckering 1.2 Broken stitch 1.3 Skipped stitch 1.4 Open seams 1.5 Dropped stitches 1.6 Uneven stitch 1.7 Twisted seam 1.8 Uncut thread 1.9 Oil mark/spot

2. Basic maintenance of sewing machine	2.1. Needle condition check 2.2. Needle change 2.3. Thread tension adjustment 2.4. Folder and guide adjustment 2.5. Oil level check
3. Productivity improvement techniques	3.1 Standard Minute Value (SMV) 3.2 Standard Operating Procedures (SOP) 3.3 Target setting 3.4 Operator efficiency 3.5 Operator capacity 3.6 Efficient Material Handling

Curricular Content Guide

1. Underpinning Knowledge	1.1 Quality specifications and standard 1.2 Product inspection procedures 1.3 Sewing faults identification technique 1.4 Basic maintenance procedure 1.5 Machine cleaning and lubricating procedures 1.6 Work schedules and plans 1.7 Productivity improvement techniques
2. Underpinning Skills	2.1 Implementing specifications and standards 2.2 Checking and verifying quality product 2.3 Identifying sewing defects and rectify 2.4 Following quality control and quality assurance 2.5 Performing basic maintenance activities 2.6 Cleaning and lubricating machine 2.7 Following work schedules and plan for productivity
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities.

	4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted quality specifications and standard 1.2 identified sewing defects 1.3 identified basic maintenance of sewing machine 1.4 cleaned and lubricated machine 1.5 carried out basic maintenance of sewing machine 1.6 understood work schedules and plans
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Advanced Sewing Machine Operation
Number of Trainees:	30

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	1
2.	Multimedia Projector	1
3.	Single Needle Lock Stitch Machine	30
4.	Overlock Machine	5
5.	Flat Lock Machine Operation	2
6.	Double Needle Lock Stitch Machine	2
7.	Feed of the Arm Machine	2
8.	Multi Needle Chain Stitch Machine	3
9.	Single Needle Chain Stitch Machine	2
10.	Bar Tack Machine	2
11.	Auto Pocket Welting (APW)	1
12.	Quality check table (Min. 5 Feet)	2
13.	Iron Set	2
14.	Iron Table	2

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Advanced Sewing Machine Operation** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 30 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 30 trainees.